BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 6 — Trigonometry and Bearing

ANSWER KEY (Version 1)

Total Marks: 65	**Time Allowed: 2h 30m**	**Version: 1**

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

1

Answer: B — 60 1 mark

By definition, 1° = 60′ (minutes). There are **3600 seconds** in a degree, not minutes. Distractor D (3600) is the seconds conversion, a common mix-up.

2

Answer: B — 2π radians 1 mark

The **circumference-to-radius ratio** of a full circle is 2π, so a complete rotation = 2π radians. π radians corresponds to only a half circle (180°).

3

Answer: C — l = rθ 1 mark

Arc length = radius × central angle in radians. This follows from the proportionality **l : θ :: r : 1**, giving l = rθ. θ must be in radians.

4

Answer: B — 1/2 1 mark

In a 30-60-90 triangle, the side opposite 30° is **half the hypotenuse**, so sin 30° = 1/2. Option A (√3/2) is sin 60°, a frequent confusion.

5

Answer: C — sinθ 1 mark

In the **second quadrant** x<0, y>0. Since sinθ = y/r with r always positive, sinθ>0. cosθ<0 (x<0) and tanθ<0 (y/x = positive/negative).

6

Answer: C — $\sin^2\theta + \cos^2\theta = 1$ 1 mark

The **fundamental Pythagorean identity**. Dividing $a^2+b^2=c^2$ by c^2 gives this. Option B should be $\sec^2\theta - \tan^2\theta = 1$ (not plus), making it wrong.

7

Answer: B — Base / Hyp 1 mark

cosθ = Base/Hypotenuse = b/c. The base is the side adjacent to angle θ and the hypotenuse is the longest side.

8

Answer: D — 50.5° 1 mark

50°30′ = 50° + (30/60)° = 50° + 0.5° = **50.5°**. This is a direct application of D°M′ to decimal conversion from the Miscellaneous Exercise.

9

Answer: B — Angle of elevation 1 mark

Angle of elevation is formed when the object is above the horizontal line of sight. Angle of depression applies when the object is below.

10

Answer: C — North 1 mark

Bearings are measured **clockwise from north** and expressed as three-figure numbers (000° to 360°). North itself has bearing 000°.

11

Answer: C — 090° 1 mark

From the Key Points: north = 000°, **east = 090°**, south = 180°, west = 270°. East is a quarter-turn clockwise from north.

12

Answer: B — 45° 1 mark

cos x = sin x implies tan x = 1, so **x = 45°**. At 45° the right triangle is isosceles (base = perpendicular), making sin and cos equal (both = $1/\sqrt{2}$). From Miscellaneous Exercise Q(v).

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) — Convert 45°45′45″ into seconds 3 marks

45°45′45″ = 45° + 45′ + 45″

= (45 × 3600)″ + (45 × 60)″ + 45″ 1 mark

= 162000″ + 2700″ + 45″ 1 mark

= 164745″ 1 mark

OR — Convert 216000″ into D°M′S″

216000″ = (216000 / 60)′ = 3600′ 1 mark

3600′ = (3600 / 60)° = 60° 1 mark

= 60°0′0″ 1 mark

2(ii) — Convert 45° into radians 3 marks

To convert degrees to radians, multiply by π/180.

45° = 45 × (π/180) radians 1 mark

= π/4 radians 1 mark

≈ 0.7854 radians 1 mark

OR — Convert 3π/4 radians into D°M′S″

3π/4 = (3/4) × (180/π)° = (3/4) × 180° = 135° 1 mark

= 135° + 0′ + 0″ 1 mark

= 135°0′0″ 1 mark

2(iii) — Arc length and area of sector: r = 12 m, θ = 120° 3 marks

Convert: θ = 120° = 120 × (π/180) = 2π/3 rad 1 mark

Arc length l = rθ = 12 × 2π/3 = 8π m ≈ 25.13 m 1 mark

Area A = $\frac{1}{2}r^2\theta$ = ½(144)(2π/3) = 48π m^2 ≈ 150.80 m^2 1 mark

OR — 30-inch pendulum, angle 30°

Convert: 30° = 30 × (π/180) = π/6 rad 1 mark

l = rθ = 30 × π/6 = 5π inches 1 mark

≈ 5 × 3.1416 ≈ 15.71 inches 1 mark

2(iv) — Six trig ratios given $\sin\theta = \sqrt{3}/2$, first quadrant 3 marks

$\sin\theta = \sqrt{3}/2$, so perpendicular = $\sqrt{3}$, hypotenuse = 2. By Pythagoras: base = $\sqrt{(4-3)} = 1$. 1 mark

$\sin\theta = \sqrt{3}/2$ | $\cos\theta = 1/2$ | $\tan\theta = \sqrt{3}$ 1 mark

$\operatorname{cosec}\theta = 2/\sqrt{3}$ | $\sec\theta = 2$ | $\cot\theta = 1/\sqrt{3}$ 1 mark

OR — Six trig ratios of 45°

At 45°: base = perpendicular = 1, hypotenuse = $\sqrt{2}$. 1 mark

$\sin 45° = 1/\sqrt{2}$, $\cos 45° = 1/\sqrt{2}$, $\tan 45° = 1$ 1 mark

$\operatorname{cosec} 45° = \sqrt{2}$, $\sec 45° = \sqrt{2}$, $\cot 45° = 1$ 1 mark

2(v) — Evaluate $\sin 60° - \cos 30°$ 3 marks

$\sin 60° = \sqrt{3}/2$ 1 mark

$\cos 30° = \sqrt{3}/2$ 1 mark

$\sin 60° - \cos 30° = \sqrt{3}/2 - \sqrt{3}/2 = 0$ 1 mark

OR — $\sec 45° \operatorname{cosec} 45° - \sin 30° \cos 60°$

$\sec 45° = \sqrt{2}$, $\operatorname{cosec} 45° = \sqrt{2}$, $\sin 30° = 1/2$, $\cos 60° = 1/2$ 1 mark

$= (\sqrt{2})(\sqrt{2}) - (1/2)(1/2) = 2 - 1/4$ 1 mark

$= 7/4 = 1.75$ 1 mark

2(vi) — $\cos\theta = 1/2$, terminal ray in first quadrant, find remaining ratios 3 marks

$\cos\theta = 1/2 \Rightarrow$ base = 1, hyp = 2. Perp = $\sqrt{(4-1)} = \sqrt{3}$. 1 mark

$\sin\theta = \sqrt{3}/2$, $\tan\theta = \sqrt{3}$ 1 mark

$\operatorname{cosec}\theta = 2/\sqrt{3}$, $\sec\theta = 2$, $\cot\theta = 1/\sqrt{3}$ 1 mark

OR — $\operatorname{cosec}\theta = 2$, second quadrant; find $\cos\theta \cdot (\sin^2\theta-\cos^2\theta)/(\sin^2\theta+\cos^2\theta)$

$\operatorname{cosec}\theta = 2 \Rightarrow \sin\theta = 1/2$. In Q2: $\cos\theta = -\sqrt{3}/2$. 1 mark

$\sin^2\theta+\cos^2\theta = 1$ (denominator). Numerator: $1/4 - 3/4 = -1/2$. 1 mark

$= (-\sqrt{3}/2) \times (-1/2) = \sqrt{3}/4$ 1 mark

2(vii) — Prove: $(1 - \sin^2\theta) \sec^2\theta = 1$ 3 marks

L.H.S. = $(1 - \sin^2\theta) \sec^2\theta$ 1 mark

Use identity: $1 - \sin^2\theta = \cos^2\theta$ 1 mark

$= \cos^2\theta \times (1/\cos^2\theta) = 1$ = R.H.S. Hence proved. 1 mark

OR — Prove: $1/(\sec\theta - \tan\theta) = \sec\theta + \tan\theta$

L.H.S. = $1/(\sec\theta - \tan\theta)$. Multiply numerator and denominator by $(\sec\theta+\tan\theta)$: 1 mark

$= (\sec\theta+\tan\theta) / (\sec^2\theta-\tan^2\theta)$ 1 mark

$= (\sec\theta+\tan\theta) / 1 = \sec\theta+\tan\theta$ = R.H.S. Hence proved. 1 mark

2(viii) — Solve triangle ABC: $\angle C=90°$, $\angle A=30°$, $a=6$ cm 3 marks

$\angle A+\angle B+\angle C=180° \Rightarrow 30°+\angle B+90°=180° \Rightarrow$ **$\angle B = 60°$** 1 mark

$\sin 30° = a/c \Rightarrow 1/2 = 6/c \Rightarrow$ **$c = 12$ cm** 1 mark

$b^2 = c^2-a^2 = 144-36 = 108 \Rightarrow b = 6\sqrt{3}$ cm 1 mark

OR — Triangle PQR: $\angle Q=90°$, $\angle P=60°$, $PQ=4\sqrt{3}$ cm

$\angle P+\angle R=90° \Rightarrow$ **$\angle R = 30°$** 1 mark

$\tan 60° = QR/PQ \Rightarrow \sqrt{3} = QR/(4\sqrt{3}) \Rightarrow$ **$QR = 12$ cm** 1 mark

$PR = PQ/\cos 60° = (4\sqrt{3})/(1/2) = 8\sqrt{3}$ cm 1 mark

2(ix) — Angle of elevation 30° from 20 m distance 3 marks

Let height of tree = h. In right triangle: $\tan 30° = h/20$ 1 mark

$1/\sqrt{3} = h/20$ 1 mark

$h = 20/\sqrt{3} = (20\sqrt{3})/3 \approx 11.55$ m 1 mark

OR — Shadow $10\sqrt{3}$ m, pole 10 m high

$\tan\alpha$ = opposite/adjacent = height/shadow = $10/(10\sqrt{3})$ 1 mark

$\tan\alpha = 1/\sqrt{3}$ 1 mark

$\alpha = 30°$ — angle of elevation of the sun 1 mark

2(x) — Definition of bearing and cardinal bearings 3 marks

Bearing: the angle measured clockwise from north, expressed as a three-digit number. 1 mark

North = 000°, East = 090° 1 mark

South = 180°, West = 270° 1 mark

OR — Bearing of A from B if bearing of B from A = 250°

Since 250° > 180°, use: bearing of A from B = bearing of B from A − 180° 1 mark

= 250° − 180° 1 mark

= 070° 1 mark

2(xi) — Car travels 040° for 13 km; find northward and eastward distances 3 marks

Bearing 040° means angle of 40° clockwise from north, so the triangle has north-component and east-component. 1 mark

Northward = $13 \cos 40° \approx 13 \times 0.766 \approx$ **9.96 km** 1 mark

Eastward = $13 \sin 40° \approx 13 \times 0.643 \approx$ 8.36 km 1 mark

OR — Plane 119 km west, 100 km south; find bearing to airport

The plane must fly north-east to reach airport. $\tan\theta = 119/100 \Rightarrow \theta$ = tan&sup-1;(1.19) ≈ 49.96° ≈ 50° (east of north) 1 mark

Since airport is to the north-east of the plane: 1 mark

Bearing = 000° + 50° = 050° (approximately) 1 mark

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i) — Arc length, sector area and central angle: r=6 dm, θ=60°45′30″ 5 marks

Step 1: Convert θ to decimal degrees. 1 mark

60°45′30″ = 60 + 45/60 + 30/3600 = 60 + 0.75 + 0.00833 = 60.7583°

Step 2: Convert to radians. 1 mark

$\theta = 60.7583 \times \pi/180 \approx 60.7583 \times 0.01745 \approx 1.0603$ rad

Step 3: Arc length $l = r\theta = 6 \times 1.0603 \approx 6.36$ dm 1 mark

Step 4: Area $A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 36 \times 1.0603 \approx 19.09$ dm² 1 mark

Verification: $A = \frac{1}{2} \times r \times l = \frac{1}{2} \times 6 \times 6.36 = 19.08$ dm² ≈ same (rounding) — verified. 1 mark

OR — l = 4 cm, A = 16 cm²; find r and θ

Step 1: From $l = r\theta$, we get $\theta = l/r = 4/r$. 1 mark

Step 2: Substitute into $A = \frac{1}{2}r^2\theta$: 1 mark

$16 = \frac{1}{2} \times r^2 \times (4/r) = 2r$

Step 3: r = 8 cm 1 mark

Step 4: $\theta = l/r = 4/8 = 0.5$ rad = **1/2 radian** 1 mark

Check: $A = \frac{1}{2} \times 64 \times 0.5 = 16$ cm² — correct. 1 mark

3(ii) — Prove two trigonometric identities **5 marks**

Identity (a): $(\sin\theta - 2\sin^3\theta) / (2\cos^3\theta - \cos\theta) = \tan\theta$ **2 marks**

L.H.S. $= \sin\theta(1 - 2\sin^2\theta) / \cos\theta(2\cos^2\theta - 1)$

Use: $1 - 2\sin^2\theta = 2\cos^2\theta - 1$ (both equal cos 2θ form; or note they are equal via $\sin^2+\cos^2=1$)

$= \sin\theta(1 - 2\sin^2\theta) / \cos\theta(1 - 2\sin^2\theta) = \sin\theta/\cos\theta =$ **$\tan\theta$ = R.H.S.** Hence proved.

Identity (b): $\sqrt{(\sec^2\theta + \operatorname{cosec}^2\theta)} = \tan\theta + \cot\theta$ **3 marks**

R.H.S. $= \tan\theta + \cot\theta = \sin\theta/\cos\theta + \cos\theta/\sin\theta = (\sin^2\theta + \cos^2\theta)/(\sin\theta\cos\theta) = 1/(\sin\theta\cos\theta)$

L.H.S. $= \sqrt{(1/\cos^2\theta + 1/\sin^2\theta)} = \sqrt{((\sin^2\theta+\cos^2\theta)/(\sin^2\theta\cos^2\theta))} = \sqrt{(1/(\sin^2\theta\cos^2\theta))} = 1/(\sin\theta\cos\theta)$

L.H.S. = R.H.S. — Hence proved.

OR — Prove two OR identities

Identity (a): $(1-\sin\theta)/(1+\sin\theta) = (\sec\theta-\tan\theta)^2$ **2 marks**

R.H.S. $= (1/\cos\theta - \sin\theta/\cos\theta)^2 = ((1-\sin\theta)/\cos\theta)^2 = (1-\sin\theta)^2/\cos^2\theta$

$= (1-\sin\theta)^2/(1-\sin^2\theta) = (1-\sin\theta)^2/((1-\sin\theta)(1+\sin\theta)) = (1-\sin\theta)/(1+\sin\theta) =$ **L.H.S.**

Identity (b): $\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta\cos^2\theta$ **3 marks**

Use $a^3+b^3 = (a+b)(a^2-ab+b^2)$ with $a=\sin^2\theta$, $b=\cos^2\theta$:

$= (\sin^2\theta+\cos^2\theta)(\sin^4\theta-\sin^2\theta\cos^2\theta+\cos^4\theta)$

$= 1 \times ((\sin^2\theta+\cos^2\theta)^2 - 3\sin^2\theta\cos^2\theta) = 1 - 3\sin^2\theta\cos^2\theta =$ **R.H.S. Hence proved.**

3(iii) — Aeroplane at 900 m altitude, angles of depression 30° and 60° **5 marks**

```
          O (Aeroplane, altitude 900 m)
         /|
        / |
       /  | 900 m
      /   |
A   /30  60\ B     C
    ------  -------|
    x           y
    (ships sailing in same direction below)
```

Step 1: Let C be the point directly below O. Let BC = y (nearer ship) and AC = x (farther ship). **1 mark**

Step 2: In right triangle OCB (angle of depression 60° means ∠OBC = 60° by alternate angles): **1 mark**

$\tan 60° = OC/BC \Rightarrow \sqrt{3} = 900/y \Rightarrow y = 900/\sqrt{3} = 300\sqrt{3} \approx 519.62$ m

Step 3: In right triangle OCA (angle of depression 30°, ∠OAC = 30°): **1 mark**

$\tan 30° = 900/x \Rightarrow 1/\sqrt{3} = 900/x \Rightarrow x = 900\sqrt{3} \approx 1558.85$ m

Step 4: Distance between ships = AB = AC − BC = x − y **1 mark**

$= 900\sqrt{3} - 300\sqrt{3} = 600\sqrt{3} \approx 1039.23$ m **1 mark**

OR — Tower 120 m, depression angles 60° and 45°

Step 1: Let T be top of tower (height 120 m). Let A and B be the two boats, B nearer. **1 mark**

Step 2: $\tan 60° = 120/TB \Rightarrow \sqrt{3} = 120/TB \Rightarrow TB = 120/\sqrt{3} = 40\sqrt{3}$ m **1 mark**

Step 3: $\tan 45° = 120/TA \Rightarrow 1 = 120/TA \Rightarrow TA = 120$ m **1 mark**

Step 4: Distance $AB = TA - TB = 120 - 40\sqrt{3}$ **1 mark**

$= 120 - 69.28 \approx 50.72$ m **1 mark**

3(iv) — Ship A to B (040°, 35 km) then B to C (130°, 50 km) **5 marks**

```
        N                  N
        |                  |
        |  040             |   130
        | /               B|-------C
        |/ 35km            |  50km
        A
```

Step 1: Angle at B between AB and BC directions. **1 mark**

The two north lines at A and B are parallel. Bearing AB = 040°, bearing BC = 130°.

Angle between AB extended and BC = 130° − 040° = 90°. So triangle ABC has a right angle at B.

Step 2 (a) — Find AC: By Pythagoras: **1 mark**

$AC = \sqrt{(AB^2 + BC^2)} = \sqrt{(35^2 + 50^2)} = \sqrt{(1225 + 2500)} = \sqrt{3725} \approx$ **61.03 km** **1 mark**

Step 3 (b) — Find bearing of C from A: **1 mark**

$\tan\alpha = BC/AB = 50/35 \approx 1.4286 \Rightarrow \alpha = \tan^{-1}(1.4286) \approx 55°$

Bearing of C from A = 040° + 055° = 095° **1 mark**

OR — P to Q (050°, 3 km), Q to R (140°), PR = 5 km

Step 1: Angle PQR: bearing PQ=050°, bearing QR=140°. Angle at Q = 140°−050°=90°. Right angle at Q. **1 mark**

Step 2 (a): $QR = \sqrt{(PR^2 - PQ^2)} = \sqrt{(25-9)} = \sqrt{16} =$ **4 km** **1 mark**

Step 3 (b) bearing of R from P: **1 mark**

$\tan\alpha = QR/PQ = 4/3 \Rightarrow \alpha = \tan^{-1}(4/3) \approx 53.13°$ **1 mark**

Bearing of R from P = 050° + 053° = 103° **1 mark**